

KAPTURE FINANCE

High-Level Design (HLD) Voice AI Collections Bot

End-to-End Architecture, State Machine, Tool Specs & Compliance Guardrails
for 'Maya'

Document Details:

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Project Name: Maya Collections Voicebot

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1. System Architecture & Pipeline

Maya is designed as a low-latency, real-time voice pipeline. Vapi.ai acts as the primary voice orchestrator, coordinating telephony, speech-to-text (STT), the LLM reasoning loop, and text-to-speech (TTS), and integrating with Kapture's webhook backend.

1.1 Media Pipeline & Latency Budgets (Per Hop)

To maintain a natural human conversation flow, the round-trip latency (Silence-to-Speech) must be < **1,200ms** (target: **800ms**).

Pipeline Step	Technology Choice	Latency Budget	Justification
Telephony	SIP / PSTN via Vapi	~100ms	Direct carrier routing for jitter reduction.
STT (Transcriber)	Deepgram Nova-2	~150ms	Fastest word-error-rate transcriber on market.
LLM Engine	OpenAI GPT-4o-mini	~250ms	Extremely fast token generation time.
TTS (Voice Synth)	Cartesia Sonic	~100ms	High emotional control, streamed synthesis.
Webhook API	FastAPI (webhook_server)	~150ms	Optimized async I/O database calls.

2. Conversation Flow & State Machine

To comply with debt collection privacy standards, the state transitions are **strictly locked** by the orchestrator state engine. The bot is strictly forbidden from disclosing any details of the debt before verifying identity.

- **State 1: Greeting (Unauthenticated)** - Hello, am I speaking with Mr. Rahul Sharma? No debt details are disclosed here. Wrong person calls terminate immediately.
- **State 2: Verification (Pending Auth)** - Ask DOB or last 4 digits of PAN. Verify via tool. Max 2 failed attempts, else hang up to protect data privacy.
- **State 3: Debt Presentation (Authenticated)** - State loan details: EMI amount ■8,499, 12 days overdue, due on Aug 5th.
- **State 4: Negotiation** - Handle customer intents (Will Pay, Cannot Pay/Hardship, Already Paid, Disputes, Callback, DNC, Abusive Caller).
- **State 5: Closure** - Summarize PTP date, send SMS payment link, mark CRM call disposition, and politely disconnect.

3. Intents & Entities Definition

Intent	Description	Entities Extracted	Action Hooked
will-pay	Customer agrees to clear payment.	ptp_date, payment_method	Log PTP date, send payment link.
cannot-pay	Financial hardship, unable to pay.	hardship_reason	Offer partial payment or escalate.
already-paid	Claims payment is already made.	payment_date, ref_number	Flag CRM for audit verification.
dispute	Disputes amount or loan validity.	dispute_reason	Log dispute, transfer to supervisor.
callback-req	Requests callback at later time.	callback_time	Schedule callback slot in CRM.
do-not-call	Requests adding to DNC list.	None	Add to DNC, block number, hang up.

4. API / Webhook Integration

The voicebot utilizes 5 structured tools to read and write database records live during a call:

1. **verify_customer(customer_id, dob, last4PAN)**: Checks DOB or PAN against records. Returns verified = true/false.
2. **log_promise_to_pay(customer_id, ptp_date, ptp_amount)**: Updates the CRM record with the committed repayment date.
3. **send_payment_link(customer_id, payment_method)**: Triggers the SMS payment link dispatch system.
4. **escalate_to_agent(customer_id, reason)**: Initiates live SIP trunk bridge transfer to a supervisor queue.
5. **mark_disposition(customer_id, disposition_code, call_notes)**: Logs the call outcome code (PTP, DNC, etc.) and brief notes on disconnect.

5. Compliance & Guardrails (RBI Fair Practice Code)

Maya is built to adhere strictly to the Reserve Bank of India (RBI) Fair Practice Code for lenders. Compliance logic is hardcoded into the conversational framework and webhook gateway:

- **Third-Party Disclosure:** Under no circumstances will Maya disclose details of a loan or overdue status to a third party (friend, colleague, relative) answering the call. Maya will immediately end the call if the debtor is unavailable, logging a 'THIRD_PARTY' disposition.
- **Permitted Calling Hours:** Outbound calling is blocked outside 8:00 AM to 7:00 PM. The gateway rejects any outbound trigger requests outside these hours.
- **Tone Guardrail:** Maya maintains a polite, professional, and empathetic tone. The prompt explicitly forbids threats, harassment, or raising voice. Abusive customers are warned, and then politely hung up on after one warning.
- **Off-Topic Redirection:** If the customer attempts to ask off-topic questions (e.g. general search questions), Maya politely redirects the conversation back to the overdue loan status.

6. Observability & Monitoring Metrics

To debug, tune, and measure Maya's performance, the dashboard and logging framework monitor several KPIs:

- **PTP (Promise to Pay) Rate:** The % of calls where the customer successfully commits to a valid payment date. Target: > 35%.
- **Containment Rate:** The % of calls resolved without escalating to a human agent. Target: > 80% on routine calls.
- **Average RTT Latency:** The time from user silence to agent voice output. Target: < 1,200ms.
- **Authentication Pass Rate:** The % of targeted contacts who complete verification successfully. Identifies wrong numbers or unreachable profiles.
- **Drop-Off Points:** Tracks the exact conversation state where customers hang up, revealing potential friction in the prompt design.